

10X COMMAND CENTER

MULTI-AGENT ORCHESTRATOR • JARVIS-GRADE UI • BUILD IN 7 DAYS

THREE LAYERS

A live, voice-controlled, multi-agent AI Employee Command Center demoed on stage at GBI Singapore 2026. Real Claude calls. Real-time streaming. Inter-agent traffic visible to the audience.

1. VISUAL SHELL

CEO node at the center, 3 department satellites orbiting (Marketing, Sales, Operations). Pulsing nodes, expanding rings, light traveling between agents, particle backdrop, glassmorphism.

React • Framer Motion • React Three Fiber • Tailwind • Arwes

2. AGENT ORCHESTRATION

A "CEO" agent routes commands to Marketing/Sales/Ops sub-agents. Each sub-agent has its own brief/voice/tools. They collaborate visibly. WebSocket streams updates as agents think.

Anthropic SDK • LangGraph supervisor • Sonnet 4.5 • CopilotKit AG-UI

3. CINEMATIC LAYER

Pre-rendered cinematic moments — boot-up intro, transition videos, ambient backdrops. Higgsfield + Seedance shine here. Not for live agent work — for the theater around it.

Higgsfield • Seedance • HTMLVideoElement • mix-blend-mode

THE TL;DR STACK

LAYER	PICK
Frontend	Next.js 15 + CopilotKit + AG-UI Protocol
Component primitives	shadcn/ui + Tailwind + Arwes (sci-fi chrome)
Motion	Motion (Framer) + GSAP timeline (boot sequence)
3D scene	react-three-fiber + drei + @react-three/postprocessing
Inter-agent lines	xyflow (React Flow) OR Magic UI AnimatedBeam
Particle backdrop	tsParticles `links` preset
Streaming text	Vercel AI SDK + flowtoken (per-token blur-in)
Audio	Howler.js sprite + Arwes BleepsProvider
Orchestration	LangGraph (Python) + langgraph-supervisor
Agent runtime	Claude Sonnet 4.5 via Anthropic SDK
Voice	OpenAI Realtime API (transcribe only) → Claude
Backdrop video	Higgsfield + Seedance • loop • screen blend

★ BUILD WINDOW

7 days, solo builder + Claude Code. ~6-8 hrs/day focused build. Days 1-5 lock in features, Days 6-7 polish + rehearsal. Don't ship anything new on Day 7.

AGENT ORCHESTRATION LAYER

These five repos give you the entire backend. Clone in this stack-rank order. Modify, don't build from scratch.

A1. CopilotKit/CopilotKit

★ 30.5k · MIT · github.com/CopilotKit/CopilotKit

Frontend protocol layer. Ships the AG-UI Protocol — streaming agent state, tool calls, intermediate steps from any backend (LangGraph, ADK, custom) into a React frontend. Adopted by Google, Oracle, LangChain. Provides `useCoAgent`, `useCoAgentStateRender`, human-in-the-loop, generative UI. **Study first:** [examples/coagents-research-canvas/](#) — full-stack LangGraph + CopilotKit canvas, mirror of demo shape. **Integration: ~4 hours baseline, ~1 day for routing visuals.**

A2. langchain-ai/langgraph + langgraph-supervisor-py

★ 30.9k · MIT · github.com/langchain-ai/langgraph

Orchestration backbone. The supervisor pattern maps 1:1 to "CEO agent → department agents". Built-in checkpointing, streaming events (`stream_mode="values"` / `"updates"` / `"messages"`), human-in-the-loop, time-travel debug. LangGraph Studio gives a free graph visualizer you can secondary-screen on stage. **Study first:** [langgraph-supervisor-py/README.md](#) — supervisor primitive in 30 lines. **Integration: ~6-10 hours.**

A3. anthropics/claude-agent-sdk-demos

★ 2.3k · MIT · TS 88% Python 9%

Anthropic's reference patterns. The [research-agent/](#) demo is a CEO-orchestrator-routes-to-subagents demo with parallel research workers. Uses `AgentDefinition`, `SubagentTracker`, `HookMatcher` — exact primitives needed to track and stream subagent activity. [simple-chat-app/](#) shows React + Express + WebSocket streaming. **Integration: ~2 hours to clone, ~1 day to fork the orchestrator pattern.**

A4. assistant-ui/assistant-ui

★ 9.8k · MIT · 200k monthly downloads

Streaming chat primitives. Streaming, auto-scroll, tool-call rendering, voice dictation, accessibility, retries — chat-UX details that eat 2 days when you build them yourself. Integrates natively with Vercel AI SDK, LangGraph, AND Mastra. Used by LangChain, Stack AI, Browser Use. Composable shadcn-style primitives. **Use it for the chat column; CopilotKit for the agent state column. Don't run both chat shells.**

A5. openai/openai-realtime-agents

★ 6.8k · MIT · Next.js + WebRTC

Voice front-end reference. The cleanest open-source pattern for browser-native voice agents with WebRTC. Includes the Chat-Supervisor pattern (small realtime model for fast voice + bigger model for reasoning) and Sequential Handoffs. **Use front-end voice but route the transcribed intent to your Claude orchestrator** — don't lock the show to OpenAI's runtime. **Critical:** use push-to-talk on stage. Open-mic VAD + audience laughter = chaos. Volunteer holds spacebar.

★ WHY NOT THE ALTERNATIVES

Not OpenAI Agents SDK as orchestrator — breaks the "Claude-powered" narrative. Use only for voice front-end.

Not CrewAI — inter-agent state is fuzzy and hard to visualize live. Audience needs to see deterministic routing.

Not AutoGen — debate-style agent loops blow latency. 4-agent debate × 5 rounds = 20+ LLM calls = 30s silence on stage.

Not Mastra — beautiful TS framework, but LangGraph + CopilotKit ecosystem has tooling Claude Code can scaffold fastest today.

JARVIS-GRADE UI LAYER

These five together give you the entire shell. Iron Man / Tron / Mission: Impossible energy. Every other repo on the next page is additive polish.

V1. arwes/arwes (@arwes/react)

★ 7.5k · MIT · github.com/arwes/arwes

This IS the Jarvis aesthetic. `Animator` system orchestrates entrance/exit states across nested components ("idle → entering → entered → exiting"). `<FrameSVGOctagon>`, `<FrameSVGCorners>` — angled chamfered-corner panels you see in every sci-fi UI. `<BleepsProvider>` for built-in UI sounds. Influences explicitly cited: Star Citizen, Halo, TRON: Legacy. **Steeper API than shadcn — budget 1 day to grok the Animator state machine.**

V2. xyflow/xyflow (React Flow)

★ 36.4k · MIT · github.com/xyflow/xyflow

Bones of the agent network. Nodes for CEO + departments, edges between them, animated edge support out of the box. Built-in `<AnimatedSVGEde>` example animates a circle along the SVG path — that's literally "data flowing between agents". **Fork:** `examples/edges/animated-svg-edge` — file `AnimatedSVGEde.tsx` is the data-flow line. **Integration: ~3 hours for radial layout.**

V3. Aceternity UI + Magic UI (copy-paste components)

★ 10k+ / 14k+ · MIT · ui.aceternity.com + magicui.design

The polish layer. Specifically copy: **Glowing Effect** (Aceternity, radial pointer-follow glow on department hover) · **Background Beams with Collision** (Aceternity, cinematic backdrop) · **Spotlight** (Aceternity, single beam highlighting CEO node on entry) · **Animated Beam** (Magic UI, fastest way to do inter-agent connections — pass refs, set `gradientStartColor` + `pathColor`, ~30 lines total) · **Terminal** (Magic UI, streaming text component) · **Marquee** (Magic UI, status ticker). **Each is single-file paste — trivial integration.**

V4. react-three-fiber + drei + react-postprocessing

★ R3F 28k, drei 9k, post 1.8k · MIT · pmndrs.org

The 3D CEO sphere with real volumetric glow. Drei: `<Sphere>`, `<MeshDistortMaterial>` (CEO core surface that subtly pulses), `<Sparkles>`, `<Float>`. Postprocessing's `<Bloom>` with `luminanceThreshold={1}` and emissive materials at `color intensity > 1` = real selective glow, not a CSS box-shadow fake. **Study:** `r3f-by-example/examples/effects/emissive-bloom/` — that's 90% of your CEO-node shader work. **Integration: 1 full day for central scene.**

V5. Vercel AI SDK + flowtoken

★ 12k+ / 528 · MIT · vercel.com/ai + ephibbs.com/flowtoken

Cinematic LLM streaming. Vercel AI SDK's `streamText` + `useChat` handles the data plumbing; flowtoken wraps each token in a `blurIn` or `fadeAndScale` animation so it doesn't look like cheap JS typewriter — it looks like the AI is *materializing* thoughts. **Fork:** `flowtoken/src/components/SmoothText.tsx` — pass `animation="blurIn"`, `animationDuration="800ms"`. **Integration: ~2 hours.**

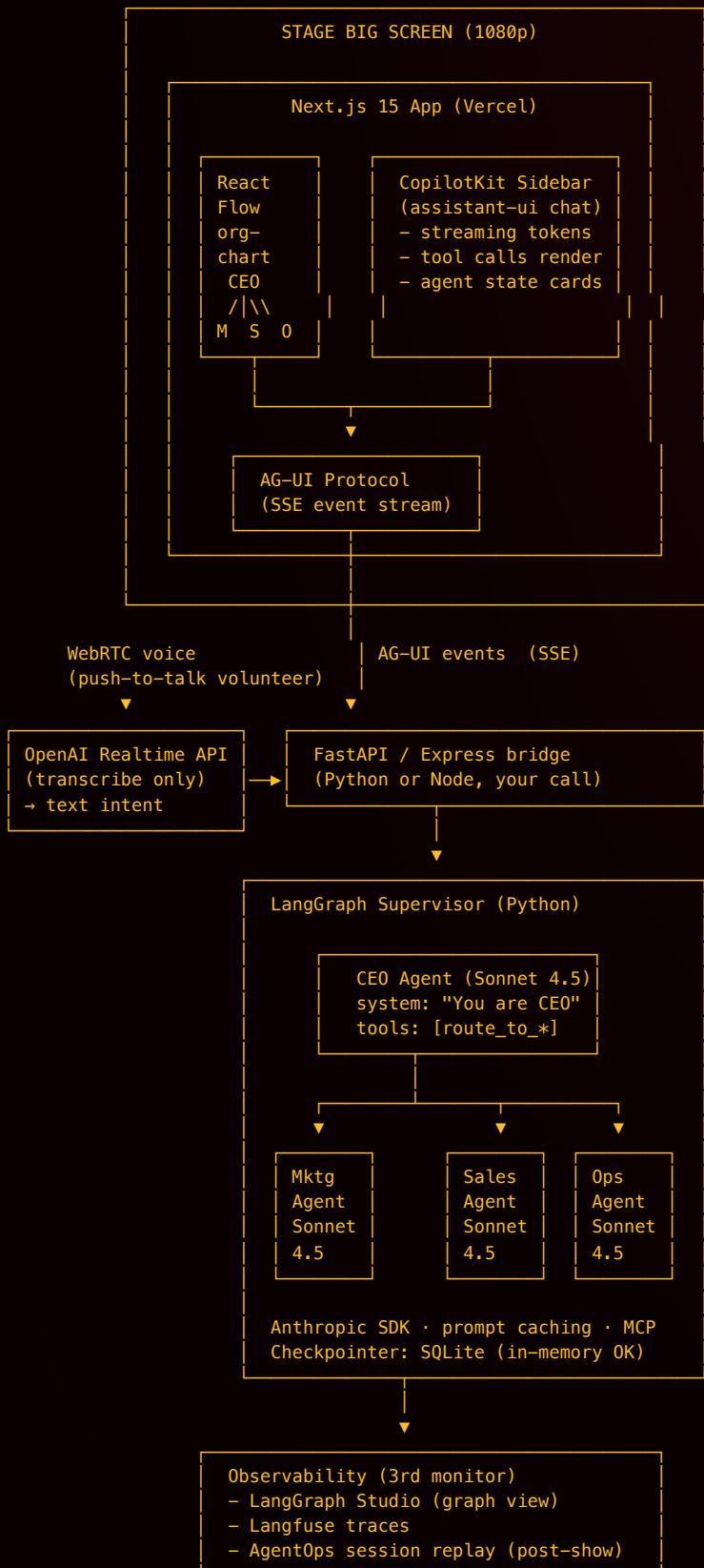
PLUS 11 SUPPORTING REPOS

REPO	USE
SciFiCN UI (baxy5/scificn-ui)	shadcn-compatible sci-fi components in 3 themes (Sci-Fi green/amber, Star Wars, Alien)

The Gridcn	Tron-styled shadcn theme, 50+ components, neon borders, scanlines
GlitchCN UI	CRT/scanline overlay — single-file paste
tsParticles `links` preset	Connecting-dots backdrop — ~80 particles, link distance 150, opacity 0.3
Recon Dashboard	Working cinematic mission-control reference (cherry-pick CRT keyframes + globe)
Howler.js	Audio sprite engine — sub-1KB triggers, zero-latency UI bleeps
r3f-particle-system	GPU particles inside CEO node — swirling vortex of light
agent-prism	AI-agent trace timeline — drop into expanded department panel
Theatre.js	After-Effects-style timeline editor for orchestrating boot sequence
CopilotKit/canvas-with-langgraph-python	Closest turnkey monorepo — clone Day 1 and modify rather than build from scratch
CopilotKit/open-multi-agent-canvas	Multi-agent-in-one-chat reference (Travel + Research + MCP)

THE SYSTEM

Voice in. Claude reasoning. Live UI streaming. Below is exactly what data flows where.



1. Volunteer holds spacebar → mic → OpenAI Realtime returns transcribed text
2. Transcribed intent POSTed to FastAPI bridge `/api/run`
3. FastAPI invokes LangGraph supervisor with the intent
4. CEO agent decides routing, calls `route_to_marketing` (or multiple in parallel)
5. Each tool call streams `update` events back via AG-UI SSE
6. CopilotKit `useCoAgentStateRender` lights up the right node in React Flow
7. Specialist agents stream their token output to the chat panel
8. CEO synthesizes and returns final response

★ TWO DAY-0 DECISIONS TO LOCK

1. **Polyglot vs. TS-only?** Polyglot (Python backend + TS frontend) wins because LangGraph Studio's graph visualizer is too valuable to give up. Recommendation: **polyglot**.
2. **Self-host CopilotKit runtime vs. Cloud?** Self-host on Vercel — one fewer external dependency on demo day. Recommendation: **self-host**.

BACKEND BUILD SPRINT

Daily target: 6-8 hours of focused build with Claude Code. Each day ends with a deliverable that's verifiable. No "code in progress."

DAY 1 · SAT · SCAFFOLD & HELLO WORLD

A running Next.js app with one Claude call streaming to a React chat.

- `git clone CopilotKit/canvas-with-langgraph-python` — rename, push to your repo
- `git clone anthropics/claude-agent-sdk-demos` — study `research-agent/` and `simple-chat-app/`
- Install: `pnpm install` frontend, `uv pip install` Python backend
- Replace canvas LLM with `ChatAnthropic(model="claude-sonnet-4-5-20250929")`
- Verify streaming: type "hello", see tokens stream
- Set up Anthropic API key + LangSmith tracing
- Read `anthropics/claude-cookbooks/patterns/agents/orchestrator_workers.ipynb` cover-to-cover

► End-to-end chat with Claude, streaming visible, LangSmith trace appears.

DAY 2 · SUN · DEFINE THE AGENTS

CEO + 3 department agents wired in LangGraph supervisor pattern.

- Install `langgraph-supervisor`
- Write 4 system prompts (CEO, Marketing, Sales, Operations) — borrow wholesale from `purely-personal:marketing-engine`, `sales-engine`, `operations-engine` skills you already have (massive shortcut)
- `create_supervisor([marketing, sales, ops], model=ChatAnthropic(...))`
- Test text-only: "Plan this week's content" should route to Marketing, return real LinkedIn drafts
- Add prompt caching to CEO system prompt (saves ~60% latency)

► Curl the FastAPI endpoint with text intent, see all 4 agents fire in trace, real content returned.

DAY 3 · MON · ORG-CHART VISUAL

React Flow with CEO + 3 department nodes that light up when active.

- Install `@xyflow/react`
- Build 4 custom node types: `<CeoNode />`, `<DeptNode />` with `idle | thinking | done` states
- Animated edges (`animated: true`) when messages flow
- Wire `useCoAgent` hook from CopilotKit to subscribe to LangGraph state updates
- Sub-agent fires → set node state to `thinking`; returns → `done`
- Polish: custom Tailwind colors per department, glow effect on active nodes

► Type "create LinkedIn content" → CEO node pulses → Marketing node lights up → returns. Audience sees the routing.

DAY 4 · TUE · DRILLDOWN + GENERATIVE UI

Click Marketing → expands to sub-categories → real work happens.

- `useCoAgentStateRender` to render generative UI cards as agents work
- Per department, define 3 sub-tools (Marketing → `draft_linkedin`, `draft_newsletter`, `draft_x_thread`)
- Expand/collapse animation in React Flow when CEO node is "clicked"
- Re-use `purely-personal:marketing-engine` prompts as Marketing agent's tool implementations

► Day 1 demo flow works end-to-end via text. "Click Marketing → Content → 'Plan this week'" produces 3 real LinkedIn posts streaming live.

DAY 5 · WED · VOICE INPUT

Push-to-talk volunteer can speak and the system reacts.

- `git clone openai/openai-realtime-agents` for reference
- Add WebRTC voice component — use `pipecat-ai/voice-ui-kit` or roll your own with Web Audio API + OpenAI Realtime
- Bind spacebar to push-to-talk; OpenAI Realtime *only* for transcription, then forward transcript to LangGraph pipeline
- **Critical:** Manual override — if voice fails on stage, type same intent in <2 seconds. Hidden Cmd+T toggle.
- Pre-script 5 stage commands and rehearse with audience-noise simulation

► Speak "create this week's LinkedIn content" → see content stream in.

DAY 6 · THU · SHOWSTOPPER · MULTI-DEPT ORCHESTRATION

"Hey team, plan this week" fires Marketing + Sales + Ops in parallel, with inter-agent visible "talking."

- In LangGraph, define meta-tool that CEO uses to dispatch to multiple departments simultaneously
- Use Claude Sonnet 4.5's parallel tool calls (fires multiple in one turn)
- Visualize parallel execution: 3 nodes glow simultaneously, edges animate
- Synthesis step where CEO node aggregates the three outputs into a unified weekly plan
- Department-color tokens stream into chat in distinct columns or with avatars

► Day 2 showstopper works. Voice → 3 departments fire in parallel → CEO synthesizes → coherent weekly business plan in <30 seconds.

DAY 7 · FRI · HARDENING + REHEARSAL

Demo-day reliable. No hot fixes.

- **Reliability:** Cache prompts, retry logic on every Anthropic call (1 retry, 1.5s backoff), pre-warm with no-op call on stage entry
- **Local fallback:** Hard-code "Plan B" canned response that triggers if any agent times out >15s. Audience never knows.
- **Rehearse:** Run full Day 1 + Day 2 demo 5× back-to-back. Time it. Each segment lands <60s of agent thinking.
- Friend makes audience noise during voice tests
- Record dry run, watch back at 2× to spot dead air
- Deploy to Vercel with stable URL. Pin SDK versions.
- Plug laptop into projector, test colors and font sizes from row 20 of fake auditorium
- Print one-page run-of-show, tape to laptop

► Demo is muscle memory. You can run it half-asleep.

VISUAL BUILD SPRINT

Run in parallel with backend (or sequentially after). Every day ends with a stage test — does it look good projected at 4K from 30 feet away?

DAY 1 · SCAFFOLD (8 HRS)

Empty stage with dark backdrop and CEO core visible. Nothing interactive.

- `npx create-next-app@latest jarvis-stage --ts --tailwind --app`
- `npm i motion @arwes/react @xyflow/react @tsparticles/react @tsparticles/preset-links three @react-three/fiber @react-three/drei @react-three/postprocessing howler ai flowtoken`
- `npx shadcn@latest init` then add `button card`
- Tailwind theme: copy SMUI's color tokens; CSS vars `--neon-cyan: #00d4ff`, `--neon-amber: #ffa500`, `--bg-void: #050810`
- `app/layout.tsx` — full-bleed dark, JetBrains Mono on body
- `components/Backdrop/ParticleField.tsx` — `tsParticles` `links` preset, opacity 0.3
- `components/Backdrop/VideoLayer.tsx` — looping video, `mix-blend-mode: screen`, opacity 0.15. Higgsfield/Seedance backdrops live here.
- `components/Scene/CEOCore.tsx` — R3F Canvas with Sphere + `MeshDistortMaterial` (emissive cyan, intensity 2) + Bloom postprocessing
- Stage test: project on TV. CEO core glow real or gradient blob?

DAY 2 · MOTION (8 HRS)

Volunteer presses node → expand → sub-panels stagger in.

- Wrap app in `<AnimatorGeneralProvider duration={{ enter: 0.6, exit: 0.4 }}>`
- `DepartmentNode.tsx` — Animator wrapper, `FrameSVGOctagon` chrome, `motion.div` for press scale
- Position 3 nodes around CEO at 120° intervals (polar math)
- `DepartmentPanel.tsx` — Motion variants with `staggerChildren: 0.08`
- `AnimatePresence` so panels animate out when collapsing
- GSAP timeline for boot sequence — CEO fades up, nodes stagger clockwise, text labels fade. Run on mount.
- Stage test: practice press 10x. Cinematic or jittery?

DAY 3 · STREAMING TEXT (6 HRS)

Each agent panel has streaming text feed when "thinking."

- `app/api/chat/route.ts` — Vercel AI SDK `streamText` endpoint to Claude
- `AgentFeed.tsx` — uses `useChat` from `ai/react`
- Wrap each token in `flowtoken` `<SmoothText animation="blurIn" animationDuration="600ms">`
- "Thinking" shimmer state before first token — Magic UI Skeleton with slow gradient sweep
- **Build a fake mode** (env flag) — pre-recorded responses played token-by-token. **You will not trust live LLM on stage.**
- Stage test: from 30ft does text read? Min font-size 24px.

DAY 4 · INTER-AGENT LINES (6 HRS)

When two agents collaborate, glowing line traces between them.

- **Path A (recommended):** Magic UI `<AnimatedBeam>` . Pass refs, set `gradientStartColor` + `pathColor` . ~30 lines total.
- **Path B (more flex):** xyflow with custom `<AnimatedSVGEde>` . Good if network graph is the WHOLE layout.
- `useAgentEventBus()` — pub/sub keyed off agent IDs; emits `("marketing", "sales", "data-flow")`
- Layer 3 beams at different opacities/widths for chunky-glow feel (single beam looks anemic)
- Stage test: trigger 5 connections at once, anything stutter?

DAY 5 · SOUND (5 HRS)

UI clicks, boot whoosh, comms blip when agents talk, ambient hum.

- Download from Zapsplat: Sci-Fi Console Beeps · UI Tones · Atmospheres
- Combine into MP3 sprite via `audiosprite` CLI
- `lib/sound.ts` — Howler instance with sprite map: `{ press: [0,320], confirm: [400,600], comms: [1100,800], whoosh: [2200,1500], hum: [4000,12000,true] }`
- `useSound("press")` hook returns play function
- Wire: hover → comms (low) · press → confirm · expand → whoosh · inter-agent → comms · background → loop hum at 0.15
- **Test on actual stage PA system, not laptop speakers.** Mix changes completely.

DAY 6 · POLISH (8 HRS)

Cut everything that looks cheesy. Run the audit.

- **3-layer depth audit** — every screen has foreground (sharp, full opacity), midground (40-60%, slight blur), background (10-20%, heavy blur)
- **Easing curves audit** — replace any default linear / ease-in-out with `cubic-bezier(0.16, 1, 0.3, 1)` (expo out)
- **Glow falloff audit** — CSS box-shadows with hard edges = cheesy. Use radial gradients or R3F bloom. If box-shadow, layer 3 at increasing blur, decreasing opacity.
- **Color temperature audit** — ONE neon (cyan), one accent (amber), grayscale rest. Two neons max.
- **Text density audit** — every dense block needs a "rest" moment. Empty space, micro-divider lines, ambient glyphs.
- Run full demo end-to-end 3 times. Find anything that breaks.

DAY 7 · STAGE REHEARSAL (FULL DAY AT VENUE)

Lock everything. Build the kill-switch. Sleep.

- Project at venue resolution. Re-test contrast at viewing distance.
- Test with house lighting at presentation level (not studio dark).
- Test audio through PA. Adjust hum + bleeps mix.
- **Build kill-switch** — single key (Esc) fades everything to black gracefully if anything goes wrong.
- **Build fake-mode toggle** that stays in fake mode no matter what — backup if internet flakes.
- Record screen capture of demo for post-event highlight reel.
- Sleep.

SOUND EFFECT SOURCING

A beep that fires just before the visual change makes the visual feel decisive. 60ms head-start makes the press feel like it caused the expansion.

ROYALTY-FREE LIBRARIES

SOURCE	BEST FOR
Zapsplat	The biggest free sci-fi UI bank. Requires free account. zapsplat.com/sound-effect-category/science-fiction/
Freesound.org	Community uploads, varied quality. CC-licensed.
Mixkit	Curated, free, no account needed for many. mixkit.co/free-sound-effects/sci-fi/
BBC Sound Effects	Free for personal/educational. Atmospheric beds. sound-effects.bbcrewind.co.uk
Pixabay Audio	Free commercial use, no attribution required.

SPECIFIC ZAPSPLAT PACKS TO DOWNLOAD DAY 5

- **Sci-Fi Console Beeps** (130 sounds) — for press / hover
- **Sci-Fi UI Tones** (113 sounds) — for confirm / select
- **Sci-Fi Computer Tones** (34 sounds)
- **Sci-Fi Atmospheres** — for ambient hum bed

SEARCH TERMS THAT FIND THE GOOD STUFF

CATEGORY	QUERIES
UI clicks	<code>interface beep clean</code> · <code>confirm select hi-tech</code> · <code>menu button futuristic</code> · <code>holographic tap</code> · <code>JARVIS interface</code> · <code>tactical UI</code>
Boot / reveal	<code>system online whoosh</code> · <code>power up sci-fi rise</code> · <code>interface activate sweep</code> · <code>holographic materialize</code> · <code>tron grid expand</code>
Comms / collab	<code>transmission start</code> · <code>data transfer beep</code> · <code>signal lock</code> · <code>comms ping</code> · <code>network handshake</code>
Ambient hum	<code>spaceship interior drone</code> · <code>sci-fi atmosphere room tone</code> · <code>command center ambient</code> · <code>cyber drone low</code>

THE EXACT 8-SOUND SPRITE

SOUND	DURATION	USE
press	200-400ms	Short clean click. Default press.
confirm	400-600ms	Slightly longer + tonal. Big actions.
hover	<200ms	Sub-200ms airy whisper, very low volume.
whoosh	800-1500ms	Panel expansion.

comms-ping	600-900ms	Agent-to-agent line fires.
boot	2-4s	Reserved for stage opener only.
error	300ms	Red-flash moment. Insurance.
hum-bed	8-15s loop	Plays at 0.12 volume the entire demo.

MIXING FOR STAGE PA • CRITICAL

SOUND	MIX LEVEL
Hum bed	-22 dB · sits UNDER spoken voice without competing
UI clicks	-8 to -10 dB · audible but never punchy
Whoosh / boot	-6 dB · theatrical moments, let them have weight

★ THE LAVALIER MIC PROBLEM

If the presenter is on a lav mic, the venue audio engineer mixes the laptop into a different bus. **Make sure they have a stem with just your demo audio** so they can ride the level live. **Test on the actual venue PA — laptop speaker mixing lies to you.**

MAKE IT LOOK REAL

These are the things that separate Tron from Kid's Show. Every rule is one specific implementation. No vague "use good design."

RULE 1

Three-layer depth — always

Every frame has 3 z-depths: Foreground (z:100, sharp, full opacity, full saturation), Midground (z:50, opacity 0.4-0.6, slight blur), Background (z:0, opacity 0.1-0.2, heavy blur, desaturated).

RULE 2

Glow falloff — never single box-shadow

Bad: `box-shadow: 0 0 20px cyan` = Geocities. Good: layer 3 stacked shadows mimicking real light bloom. Better: R3F Bloom on emissive material with intensity >1.

RULE 3

Easing curves — never linear, never default ease-in-out

Default UI: `cubic-bezier(0.16, 1, 0.3, 1)` (expo out). Critical moments: `cubic-bezier(0.87, 0, 0.13, 1)` (expo in-out, "fluid intelligence"). Never Tailwind's `transition-all`.

RULE 4

Opacity rhythm — never 100% on text below the fold

Active panel headline 100%, body 80%, labels 60%. Adjacent panel everything 50% x those values. Breathable hierarchy without changing color.

RULE 5

The Ambient Text rule (Jayse Hansen's secret)

Iron Man HUD looks alive because of UNREADABLE micro-text everywhere. Ring of tiny rotating coordinates, numerical readouts updating every 200ms with random-but-bounded values, glyphs at periphery. None means anything. Sells "this is alive."

RULE 6

Every interactive element has 3 states + hover bleed

Idle (60% opacity, soft glow), Hover (100%, brighter glow, +1% scale, soft comms sound at 0.2 vol), Active (100%, full glow, slight rotation 1deg, confirm sound). Press → release has 200ms decay where glow dims back.

RULE 7

Neon palette discipline

Pick exactly 2 neons. Cyan #00d4ff for "alive", Amber #ffa500 for "alert/active". Everything else grayscale. Third neon = Christmas tree. Brief red flash for error only as momentary state.

RULE 8

Frame chrome is structural, not decorative

Use Arwes `<FrameSVGOctagon>` or hand-built SVGs. The 45° chamfer at top-right and bottom-left reads as "engineered," not "rectangular box." Without this, your panels look like Bootstrap cards.

RULE 9

The 1.5x animation rule

Motion that feels right at 1.0x in dev tools feels rushed on stage. Build with `globalMotion = 1.5` multiplier and tune. Audience needs reaction time.

RULE 10

Negative space audit

At any moment, ≥40% of screen is dark/empty. If full of panels, cut. Blade Runner 2049 has shots where 80% of frame is black. Eye needs rest to register magic.

RULE 11

Sub-pixel grid alignment

Every element on a 4px grid (Tailwind defaults match). Misaligned by 2px = amateur dashboard. Use `grid` and `gap-4` everywhere.

RULE 12

Don't animate everything at once

Stagger. 3 panels appear at 0ms, 150ms, 300ms — not all together. Apply to every group of >1 element. Motion's `staggerChildren` makes this trivial.

RULE 13

Sound timing locks visual

Bleep that fires JUST BEFORE visual change makes change feel decisive. `playSound('confirm');`

RULE 14

Backdrop video tricks

Higgsfield/Seedance clips with `mix-blend-mode: screen` or `lighten` and opacity 0.12-0.18 lets cinematic

`setTimeout(triggerExpand, 60)` . 60ms head-start makes press feel like it CAUSED expansion.

energy bleed through without competing with UI. NEVER full opacity behind UI — eats contrast.

WHAT WILL GO WRONG

Plan for failure. Have a recovery for every plausible disaster. The kill-switch has saved more demos than any retry logic.

RISK	LIKELIHOOD	IMPACT	MITIGATION
Voice fails in noisy room	High	Show-stopper	Push-to-talk only. Pre-test with 80dB noise loop. Hidden Cmd+T fallback. Don't use open-mic VAD on stage.
Anthropic API rate-limit / outage	Medium	Show-stopper	Pre-cache prompts. Recorded video fallback for each segment. Tier 4+ account.
Inter-agent latency >20s	Medium	Pacing kill	Sonnet 4.5 (2× faster than 4.0). Haiku 4.5 for routing if too slow. <code>parallel_tool_calls</code> . Pre-warm with no-op call on entry.
WebSocket / SSE drops mid-demo	Low	Visible glitch	Vercel HTTP/2 SSE. Test on hotel WiFi. Hotspot backup.
CopilotKit + assistant-ui collision	Medium	Build delay	Pick ONE chat shell on Day 1. Don't run both.
LangGraph supervisor loops / hallucinates routing	Low-Med	Wrong department fires	Add <code>routing_confidence</code> field. If <0.7, default to "general" + ask. Pre-script 3 demo intents, bake routing via few-shot.
Agent produces bad/embarrassing content live	Low-Med	Brand risk	Content validator step (Haiku checks length + no profanity). Pre-script demo prompts to known-good output range.
Over-engineering visualization	High	Time sink	Day 3 boxed at 1 day. If React Flow is fighting you by lunch, fall back to CSS Grid + motion-react and ship. Audience cares about story, not prettiness.
Vendor lock-in to OpenAI for voice	Low	Narrative ding	Voice is just transcription, not reasoning. Pitch is "Claude reasoning, OpenAI ears." Swap to Pipecat + Deepgram in 1 day post-show if needed.
Polyglot deployment friction	Medium	Day 7 hassle	Vercel for frontend, Railway/Fly for Python backend. OR use AI SDK's TS-native ToolLoopAgent (loses LangGraph Studio).

★ THE KILL-SWITCH (BUILD THIS DAY 7)

Bind `Esc` to a state that fades all panels out and shows just the CEO core with one line: **"SYSTEM IDLE"**. If anything goes wrong on stage, you press Esc, the visual stays intentional, and you say *"Let's pause here."*

This has saved more demos than any retry logic.

★ THE FAKE-MODE TOGGLE

Build both modes. **Default to fake mode for stage** (deterministic, instant, no API failures). Use real mode in green room to record live demo for highlight reel.

THE DAY-0 SETUP

Run these commands. In this order. Today.

REPO CLONE ORDER (STACK RANK)

```
# 1. The monorepo skeleton
git clone https://github.com/CopilotKit/canvas-with-langgraph-python.git jarvis-stage
cd jarvis-stage

# 2. The orchestrator pattern (study research-agent/)
git clone https://github.com/anthropics/claude-agent-sdk-demos.git ../reference-anthropic

# 3. The multi-agent UI pattern
git clone https://github.com/CopilotKit/open-multi-agent-canvas.git ../reference-multiagent

# 4. READ patterns/agents/ before writing code
git clone https://github.com/anthropics/claude-cookbooks.git ../reference-cookbooks

# 5. Voice front-end reference (Day 5)
git clone https://github.com/openai/openai-realtime-agents.git ../reference-voice
```

FRONTEND INSTALL (VISUAL STACK)

```
# Inside jarvis-stage/
npm i motion @arwes/react @xyflow/react \
  @tsparticles/react @tsparticles/preset-links \
  three @react-three/fiber @react-three/drei \
  @react-three/postprocessing \
  howler ai flowtoken

# shadcn primitives
npx shadcn@latest init
npx shadcn@latest add button card

# Optional: Aceternity + Magic UI components (copy-paste, not npm)
# Visit ui.aceternity.com and magicui.design – copy specific files.
```

BACKEND INSTALL (AGENT STACK)

```
# Python backend (use uv for speed)
cd backend/
uv pip install langgraph langgraph-supervisor \
  anthropic langchain-anthropic \
  fastapi uvicorn

# Set API keys
export ANTHROPIC_API_KEY="sk-ant-..."
export LANGCHAIN_API_KEY="ls_..." # optional, for LangSmith tracing
export LANGCHAIN_TRACING_V2="true"
```

THE TWO DAY-0 DECISIONS

DECISION	RECOMMENDATION
Polyglot vs. TS-only?	Polyglot. LangGraph Studio's graph visualizer alone justifies it.
Self-host CopilotKit vs. Cloud?	Self-host on Vercel. One fewer external dependency on demo day.

ALREADY IN YOUR FOLDER · REFERENCE MATERIAL

FILE	PURPOSE
research/03-jarvis-build-stack.md	Full backend research (33KB) — read for nuance
research/04-jarvis-visual-stack.md	Full visual research (33KB) — read for nuance
research/05-jarvis-demo-flow.md	The exact demo flow on stage
research/07-execution-plan.md	Execution plan synthesizing all research
research/08-master-plan.md	Master plan (older iteration)
research/09-master-plan.md	Master plan (latest)
research/10-build-hollywood.md	Hollywood polish techniques
research/11-hollywood-polish.md	Additional polish
research/12-the-100x-additions.md	100× additions
research/13-1000x-repos.md	1000× repos

★ CLOSING NOTE

The presenter is going to look like a magician. **Make sure the magic is in the rehearsal, not the code.**

Days 1-5 lock in features. Days 6-7 are dedicated polish. Don't ship anything new on Day 7. The biggest risk is scope creep on Day 4 (inter-agent connections) — if not landing by end of Day 4, ship Path A (Magic UI AnimatedBeam) and move on. Don't fork xyflow at midnight on Day 5.